

Does Portfolio Optimization Add Value?

An Out-of-Sample Analysis of Estimation Risk, Market Regimes, and Transaction Costs

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APPLICATION INTEGRITY NOTE

This manuscript is finalized as an academically defensible research design and application-ready research document. It deliberately contains no invented empirical findings. A completed empirical paper requires one final execution of the reproducibility pipeline against a documented historical data snapshot; numerical results must then replace the results protocol. Present the project as Independent Research, not as a publication.

1. Abstract

This independent research project investigates whether portfolio optimization adds value relative to simple diversification once estimation risk, market regimes, turnover, and transaction costs are incorporated into a genuinely out-of-sample investment process. Four portfolio-construction rules are compared: equal weight, global minimum variance, mean-variance utility optimization, and equal risk contribution. The study uses a diversified seven-ETF universe spanning U.S. equities, developed and emerging international equities, long-duration U.S. Treasuries, gold, broad commodities, and U.S. real estate. Portfolio weights are estimated recursively using only information available before each investment month. The baseline specification uses a 60-month rolling estimation window, monthly rebalancing, long-only fully invested portfolios, a 30% asset-level weight cap, and 15 basis points of proportional transaction costs. Robustness analysis varies the estimation window, transaction costs, covariance estimator, rebalancing frequency, and mean-variance risk-aversion parameter. Performance is evaluated using return, volatility, Sharpe and Sortino ratios, drawdown, VaR, CVaR, skewness, kurtosis, terminal wealth, turnover, concentration, and realized risk contributions. Moving-block bootstrap procedures are used for uncertainty intervals, while factor regressions with heteroskedasticity-and-autocorrelation-consistent inference are used where compatible factor data are available. The central contribution is methodological: the study treats implementation frictions and model risk as part of the investment problem rather than as afterthoughts.

2. Introduction

Modern portfolio theory provides a formal framework for choosing portfolio weights by balancing expected return and risk. The framework is elegant in theory but difficult to implement because the quantities required by the optimizer—particularly expected returns and covariances—must be estimated from finite samples. Estimation error can therefore translate directly into unstable weights and weak out-of-sample performance.

This creates an important empirical question for asset management: does a more sophisticated optimizer actually improve investment outcomes relative to a transparent 1/N allocation once the optimizer is exposed to realistic information constraints and trading frictions? Equal weighting is a demanding benchmark because it requires no return forecasts, has low model complexity, and is naturally diversified.

The project addresses this question using a strict rolling out-of-sample design. At each portfolio formation date, only historical information available before the investment month is used to estimate portfolio parameters. The resulting weights are then applied to the next month. Gross and transaction-cost-adjusted performance are reported separately so that statistical allocation efficiency can be distinguished from implementation value.

The project is deliberately designed as an empirical-finance study rather than a claim that one portfolio rule is universally optimal. The objective is to identify when optimization appears to add value, when that value disappears under realistic assumptions, and how sensitive conclusions are to estimation choices, market regimes, and implementation costs.

3. Research Question and Hypotheses

Primary research question: Does portfolio optimization add value once estimation risk, market regimes, turnover, and transaction costs are incorporated into a genuinely out-of-sample investment process?

RQ1: Do optimized portfolios deliver higher out-of-sample risk-adjusted performance than equal weight?

RQ2: Do minimum-variance and equal-risk-contribution portfolios provide superior downside-risk control?

RQ3: Are strategy rankings stable across bull, neutral, and bear market regimes?

RQ4: How sensitive are conclusions to estimation-window length, covariance estimation, rebalancing frequency, and transaction costs?

RQ5: To what extent can observed performance differences be explained by systematic factor exposures?

H1: Optimization strategies will not uniformly dominate equal weighting out of sample.

H2: Covariance shrinkage will improve the stability of optimized portfolios, particularly when estimation error is material.

H3: Transaction costs will reduce the apparent advantage of higher-turnover strategies.

H4: Relative strategy performance will vary across market regimes.

H5: Any economically meaningful advantage should survive reasonable changes in estimation window, rebalancing frequency, transaction costs, covariance estimation, and risk-aversion parameters to be classified as robust.

4. Literature Review

Markowitz (1952) established the mean-variance framework in which portfolio selection depends on expected returns, variances, and covariances. The framework remains the theoretical foundation for the minimum-variance and mean-variance rules evaluated in this study.

DeMiguel, Garlappi, and Uppal (2009) demonstrated the importance of estimation error by comparing optimized strategies with the naive 1/N benchmark. Their results motivate a design in which optimization is evaluated using realized out-of-sample performance rather than in-sample objective values.

Risk-parity methods provide an alternative that reduces dependence on expected-return forecasts by seeking a more balanced distribution of portfolio risk contributions. Maillard, Roncalli, and Teiletche (2010) formalize the properties of equal-risk-contribution portfolios.

Covariance estimation is a central source of model risk. Ledoit and Wolf (2004) show that shrinkage can improve covariance estimation by trading some sample information for greater stability. This study therefore treats shrinkage as a pre-specified robustness specification rather than selecting whichever covariance estimator produces the most favorable historical result.

Transaction costs are economically important because optimization can generate unstable or concentrated weights that require frequent trading. The empirical design therefore reports both gross and net performance and treats turnover as a first-class implementation diagnostic.

5. Data and Asset Universe

The baseline universe consists of seven liquid exchange-traded funds: SPY, EFA, EEM, TLT, GLD, DBC, and VNQ. The universe is intentionally small and transparent so that portfolio construction, diagnostics, and reproducibility can be inspected directly.

SPY represents U.S. equities; EFA developed-market equities outside the United States and Canada; EEM emerging-market equities; TLT long-duration U.S. Treasuries; GLD gold; DBC broad commodities; and VNQ U.S. real estate. The combination provides economically distinct risk exposures while remaining sufficiently compact for a transparent empirical study.

Historical prices are retrieved programmatically from Yahoo Finance through yfinance. Adjusted prices are used to incorporate distributions and splits. Monthly total returns are constructed from the adjusted-price series. The exact retrieval date, sample start and end dates, ticker list, missing observations, and package versions must be stored with the executed output package.

The common sample begins in 2007 so that all seven instruments can be evaluated over a broadly comparable history. The initial estimation window is excluded from the out-of-sample performance period. The final sample end date is determined by the actual data retrieval and recorded in the execution manifest.

6. Baseline Empirical Design

The baseline specification is fixed before empirical results are interpreted: 60-month rolling estimation window, monthly rebalancing, 15 basis points of proportional transaction costs, long-only fully invested portfolios, and a 30% maximum

weight per asset. No leverage or short selling is permitted.

Equal weight assigns $1/N$ to every asset and serves as the principal benchmark because it requires no parameter estimation beyond the number of assets.

Global minimum variance solves $\min w' \Sigma w$ subject to $1'w=1$ and $0 \leq w_i \leq 0.30$, where Σ is estimated from the rolling training window.

Mean-variance optimization maximizes $\mu'w - (\gamma/2)w' \Sigma w$ subject to the same constraints. The baseline risk-aversion parameter is $\gamma=3$. Sensitivity analysis evaluates $\gamma \in \{1, 3, 5, 10\}$. Because expected-return estimates are especially noisy, this strategy is treated as the most estimation-sensitive rule.

Equal risk contribution seeks weights whose component contributions to portfolio volatility are approximately equal. The final implementation should use constrained nonlinear optimization rather than a heuristic iteration so that the risk-parity definition is explicit and auditable.

At month t , portfolio parameters are estimated using only the preceding W months. The weights generated at t are applied to the subsequent investment month. The same rebalancing schedule is applied to all strategies so that trading frequency is not confounded with allocation methodology.

7. Transaction Costs and Implementability

Turnover is defined as the one-way L1 change in portfolio weights between consecutive rebalancing dates. The baseline proportional transaction cost is 15 basis points per unit of turnover. Sensitivity analysis uses 0, 5, 10, 15, and 30 basis points.

Gross returns measure the theoretical performance of the allocation rules before trading costs. Net returns subtract transaction costs based on the realized turnover. This distinction is central to the research question because a strategy can exhibit a gross advantage that is economically irrelevant after implementation frictions.

The study also reports average turnover, maximum observed asset weight, effective number of holdings, weight dispersion, and the frequency with which the 30% weight cap binds. For equal-risk-contribution portfolios, realized component-risk shares are reported to verify that the intended risk allocation is actually achieved.

8. Performance Measurement and Statistical Inference

Primary performance metrics are annualized return, annualized volatility, Sharpe ratio, Sortino ratio, maximum drawdown, 5% historical VaR, 5% CVaR, skewness, kurtosis, terminal wealth, and turnover. Both gross and net versions are retained where economically meaningful.

Sharpe ratios are calculated using a zero-risk-free-rate convention for the primary comparison and a 2% annual risk-free-rate sensitivity. The 2% value is a sensitivity anchor, not a claim about the realized risk-free rate throughout the historical sample.

Because monthly strategy returns can exhibit serial dependence, moving-block bootstrap procedures are used for uncertainty intervals. Pairwise strategy comparisons are reported as confidence intervals for performance differences rather than relying solely on point estimates.

Where factor data are available and correctly aligned, strategy excess returns are regressed on Fama-French factors and momentum. Heteroskedasticity-and-autocorrelation-consistent standard errors are used. A return advantage that disappears after factor controls is not described as alpha.

Multiple-testing risk is acknowledged explicitly. The baseline specification is fixed before interpretation, and robustness specifications are treated as diagnostics rather than as opportunities to select the most favorable historical result.

9. Market-Regime Analysis

Regimes are defined ex ante using information available before the evaluated month. The baseline rule classifies a month as Bull when the trailing 12-month SPY total return exceeds +10%, Bear when it is below -10%, and Neutral otherwise.

The purpose of regime analysis is not to search retrospectively for periods that favor a selected strategy. Instead, the fixed classification tests whether strategy rankings are conditional on broad market conditions.

Regime-specific results are reported only when each regime contains a sufficient number of observations for meaningful interpretation. Small-sample regime results are flagged as descriptive rather than treated as statistically

decisive evidence.

10. Robustness and Model-Risk Tests

Estimation-window robustness compares 36-, 60-, and 120-month rolling windows. A conclusion that reverses under modest changes in window length is classified as specification-sensitive.

Covariance robustness compares the sample covariance estimator with Ledoit-Wolf shrinkage. The objective is to determine whether apparent optimization benefits depend on a fragile covariance estimate.

Rebalancing robustness compares monthly with quarterly rebalancing. Transaction-cost robustness evaluates 0, 5, 10, 15, and 30 basis points. Mean-variance robustness varies γ across 1, 3, 5, and 10.

The study also evaluates concentration, turnover, effective holdings, weight dispersion, weight-cap binding frequency, and risk-contribution dispersion. These diagnostics help distinguish genuine diversification benefits from performance generated by concentrated or unstable allocations.

11. Results Protocol and Research Integrity

The empirical results section is intentionally generated from the reproducibility pipeline rather than manually written from memory or expectation. The final manuscript must contain the actual executed sample dates, strategy returns, risk metrics, bootstrap intervals, robustness results, and factor-regression estimates.

No performance statistic, ranking, p-value, confidence interval, or economic conclusion should be inserted unless it can be traced to an output generated by the executed code. If a data source changes, the execution date and data snapshot must be recorded.

The completed application package should preserve the code, requirements file, execution manifest, raw or legally redistributable data snapshot where permitted, generated CSV tables, generated figures, and final manuscript. This makes the project auditable by an informed reader.

Importantly, the project should be described as independent research, not as published or peer-reviewed research, unless external publication or review actually occurs.

12. Limitations

The asset universe is intentionally small and ETF-based, so the findings should not be generalized automatically to individual securities, institutional portfolios, or alternative datasets.

Transaction costs are stylized proportional costs and do not model security-specific bid-ask spreads, market impact, taxes, financing costs, or liquidity constraints. Monthly data also abstract from intramonth execution.

The mean-variance rule depends on estimated expected returns and is therefore inherently vulnerable to parameter error. Regime definitions are deliberately simple and depend on the selected threshold.

Historical backtests cannot establish that future performance will replicate historical behavior. The purpose of the study is to evaluate robustness within the stated historical experiment, not to make an investment recommendation.

13. Conclusion

The study is designed to answer a practical empirical-finance question: whether portfolio optimization creates value after the estimation and implementation problems that matter in real portfolio construction are incorporated explicitly. The benchmark is deliberately demanding because equal weighting is transparent, diversified, and free of return-forecast estimation.

The final conclusion must be determined by the complete empirical evidence. An optimized strategy should be described as superior only if its advantage is economically meaningful, survives transaction-cost adjustment, and remains reasonably stable across the pre-specified robustness tests. If an advantage disappears under realistic costs or changes materially with estimation choices, the appropriate conclusion is that optimization is conditional rather than universally superior.

This design therefore treats model risk, implementation friction, and statistical uncertainty as part of the portfolio-construction problem. That framing is the principal research contribution of the project.

14. References

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Appendix A. Pre-Specified Baseline

Specification	Baseline
Universe	SPY, EFA, EEM, TLT, GLD, DBC, VNQ
Return frequency	Monthly total returns from adjusted prices
Estimation window	60 months
Rebalancing	Monthly
Portfolio constraints	Long-only; fully invested; max 30% per asset; no leverage
Transaction cost	15 bps baseline; 0/5/10/15/30 bps sensitivity
Strategies	Equal weight; minimum variance; mean variance; equal risk contribution
MV risk aversion	$\gamma = 3$ baseline; 1/3/5/10 sensitivity
Covariance robustness	Sample covariance vs Ledoit-Wolf shrinkage
Window robustness	36/60/120 months
Regime rule	Trailing 12-month SPY return: Bull > +10%; Bear < −10%; Neutral otherwise
Inference	Moving-block bootstrap; HAC/Newey-West factor regression where applicable

Appendix B. CV-Ready Description

Independent Research — Portfolio Optimization & Risk Management | 2026 Designed a reproducible out-of-sample empirical study comparing equal-weight, minimum-variance, mean-variance, and equal-risk-contribution portfolios across a diversified multi-asset ETF universe. Applied rolling-window optimization, covariance estimation and shrinkage, transaction-cost modelling, downside-risk analysis, regime analysis, bootstrap inference, factor attribution, and portfolio-implementation diagnostics in Python.

Appendix C. Application Positioning

For MSc Finance applications, present this under Independent Research or Research Experience. Do not describe it as published, peer-reviewed, supervised, or institutionally affiliated unless that is factually true. The strongest application signal comes from being able to explain the research question, why equal weight is a demanding benchmark, how look-ahead bias is prevented, why transaction costs matter, and how robustness tests distinguish stable evidence from specification-dependent historical performance.